Assignment No. 2

Title: Implementation of Backpropagation Neural Network for Student Performance Prediction

Problem Statement :

To design and implement an Artificial Neural Network (ANN) using the Backpropagation learning algorithm to analyze student behavioral, academic, and lifestyle parameters and predict their exam performance score. The system aims to learn hidden relationships among multiple influencing factors and produce accurate regression predictions.

Objectives

1. To understand the concept of Artificial Neural Networks (ANN) and the Backpropagation learning algorithm.
2. To design and implement a multi-layer neural network model for predicting student exam performance.
3. To analyze the relationship between academic, behavioral, and lifestyle factors and student performance.
4. To train the neural network using forward propagation and backpropagation techniques.
5. To evaluate the model performance using regression metrics such as Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2 Score.
6. To visualize the training process using loss graphs and prediction comparison plots.

Theory :

Artificial Neural Networks are computational models inspired by biological neural systems. They consist of interconnected processing units called neurons arranged in layers — input, hidden, and output layers.

Backpropagation is a supervised learning algorithm widely used to train multi-layer neural networks. It minimizes prediction error by propagating the output error backward through the network and adjusting connection weights accordingly.

In this assignment, student attributes such as study hours, sleep patterns, mental health, productivity, and digital usage are used as input features. The neural network learns nonlinear relationships between these attributes and exam scores.

The training process includes:

- Forward propagation
- Loss calculation
- Gradient computation
- Backward error propagation
- Weight optimization (Adam optimizer)

Backpropagation improves prediction accuracy iteratively over multiple epochs.

Dataset Description

The dataset used in this assignment contains information related to student academic behavior, lifestyle habits, and study patterns. The purpose of the dataset is to analyze how different factors influence student exam performance.

The dataset consists of multiple input features representing different aspects of student life, such as:

- Study hours per day
- Sleep duration
- Mental health level
- Productivity score
- Digital or social media usage
- Class participation
- Physical activity
- Previous academic performance

These attributes are used as input features for the neural network model.

Dataset Characteristics

Attribute	**Description**
Dataset Type	Educational Performance Dataset
Input Features	Student behavioral and academic parameters
Output Variable	Exam Performance Score
Learning Type	Supervised Learning
Prediction Type	Regression

The dataset is divided into training and testing sets so that the neural network can learn patterns from the training data and then evaluate its prediction performance on unseen data.

Before training the neural network, the dataset is preprocessed and normalized to ensure that all input features are in a comparable scale. This helps improve the learning efficiency of the backpropagation algorithm.

The trained neural network then learns hidden relationships between these student attributes and their exam performance, allowing the model to accurately predict exam scores based on the provided input parameters.

Algorithm — Backpropagation :

Stepwise Procedure

Step 1: Initialize network weights and biases randomly.

Step 2: Input training dataset into the network.

Step 3: Perform Forward Propagation

- Compute weighted sum
- Apply activation functions

Step 4: Generate predicted output.

Step 5: Compute loss using Mean Squared Error (MSE).

Step 6: Calculate output layer error.

Step 7: Propagate error backward to hidden layers.

Step 8: Compute gradients using chain rule.

Step 9: Update weights:

$$W_{new} = W_{old} - \eta \, (\partial Loss / \partial W)$$

Step 10: Repeat steps for all epochs until loss is minimized.

System Architecture :

The implemented ANN architecture consists of multiple dense layers.

Architecture Layers

- **Input Layer:** Student feature parameters
- **Hidden Layer 1:** 128 neurons — ReLU
- **Hidden Layer 2:** 64 neurons — ReLU
- **Hidden Layer 3:** 32 neurons — ReLU
- **Output Layer:** 1 neuron — Linear activation

Architecture Flow :

Student Dataset

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Data Preprocessing

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Input Layer

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Hidden Layer 1 (128)

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Hidden Layer 2 (64)

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Hidden Layer 3 (32)

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Output Layer (Exam Score)

Comparative Result :

In this assignment, model performance evaluation was conducted after training the Artificial Neural Network using the Backpropagation algorithm.

A direct numerical comparison with "Before ANN Training" models was not performed, as no baseline machine learning or statistical model was implemented prior to ANN training.

Instead, the effectiveness of the neural network was assessed using training and validation performance metrics such as loss reduction, prediction accuracy, and regression evaluation scores.

The comparative analysis was therefore based on:

- Training vs Validation performance
- Loss minimization across epochs
- Prediction accuracy on test dataset

Comparative Analysis — Training Perspective

Metric	Value
Mean Squared Error (MSE)	34.12
Mean Absolute Error (MAE)	4.53
R^2 Score	0.75
Training Loss (Final Epoch)	15.78
Validation Loss (Final Epoch)	33.17

Result — Graphical Representation

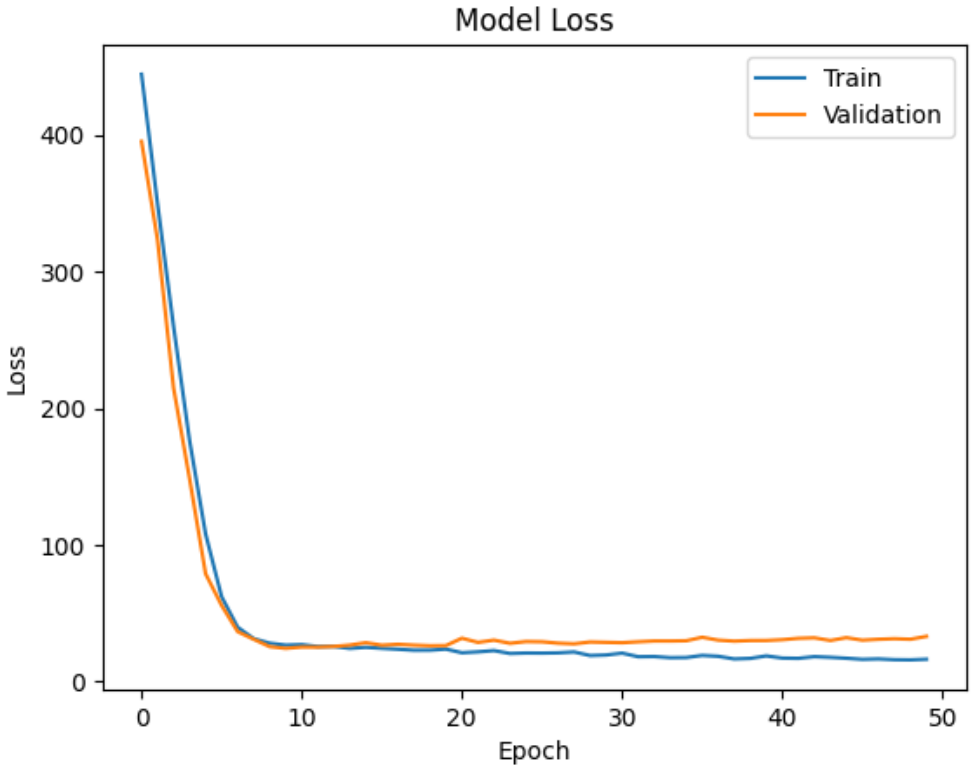
The model's learning performance was visualized using graphs plotted during ANN training.

Graph 1 — Training vs Validation Loss

Description:

- X-axis → Epochs
- Y-axis → Loss (MSE)
- Training loss decreases continuously
- Validation loss follows similar pattern
- Indicates effective backpropagation learning

This confirms that the neural network successfully minimized prediction error over training iterations.

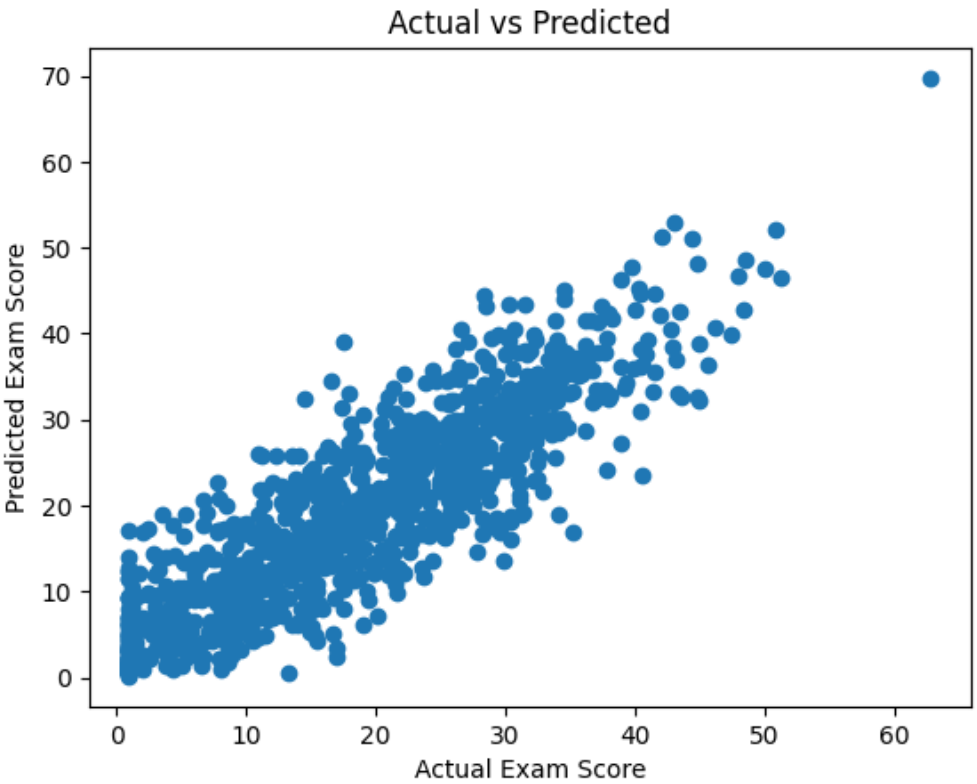


Graph 2 — Actual vs Predicted Scores

Description:

- Scatter plot comparing predicted and actual exam scores
- Points distributed near diagonal reference line
- Shows strong regression fit

This indicates the ANN model's capability to accurately estimate student performance.



Conclusion

The Backpropagation Neural Network was successfully implemented to predict student exam performance using academic and behavioral parameters.

The model demonstrated effective learning through steady loss reduction and accurate regression predictions. Training and validation results confirmed proper convergence without overfitting.

This assignment validates the applicability of Artificial Neural Networks in educational data analysis, enabling performance prediction and data-driven academic support systems.

References

1. Ian Goodfellow — *Deep Learning*, MIT Press
2. Simon Haykin — *Neural Networks and Learning Machines*
3. Christopher Bishop — *Pattern Recognition and Machine Learning*
4. TensorFlow Keras Documentation
5. Scikit-Learn Documentation
6. Kaggle — Student Performance Dataset